

SCI225 Week 4 — Complete Study Guide

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Neurologic Structure & Function · Pain, Temperature, Sleep & Sensory · Cognition · Cerebral Hemodynamics · Motor Function

Study Spine — for every neuro topic ask:

1. CNS or PNS?
2. Which cell or pathway is broken?
3. What's the location/level of the lesion?
4. UMN sign or LMN sign?

MECHANISM

WHY

COMPENSATION

PATHOLOGY

MISSTEP

ACRONYM

EXAM TIP

Course alignment

This guide maps to the three Week 4 learning objectives:

- **LO1** — Identify Structure and Function of the Neurologic System (Units 1, 2)
- **LO2** — Explain Pain, Temperature, Sleep, and Sensory Function (Units 3, 4)
- **LO3** — Discuss Alterations in Cognitive Systems, Cerebral Hemodynamics, and Motor Function (Units 5, 6, 7)

Every practice question here was written fresh to teach you to apply the concept. None reproduces an exam item.

Unit 1 • Cells of the Nervous System & Conduction LO1

1.1 Neurons and Neuroglia (Support Cells)

MECHANISM A **neuron** has three parts: dendrites (receive), cell body/soma (process), axon (transmit). The axon is wrapped in **myelin**, which insulates it and lets the action potential *jump* from one node of Ranvier to the next — this is **saltatory conduction**. Myelinated axons conduct ~50× faster than unmyelinated ones.

ACRONYM / MNEMONIC The 6 neuroglial (support) cell types — "**MOSES + AM**": Microglia · Oligodendrocytes · Schwann · Ependymal · Satellite · Astrocytes · plus Many neurons supported by all of them.
4 are CNS (astrocytes, oligodendrocytes, microglia, ependymal). 2 are PNS (Schwann, satellite).

Cell	Location	Function	If damaged...
Astrocytes	CNS	Form the blood-brain barrier; potassium buffering; metabolic support	Toxins cross BBB → neuroinflammation, edema
Oligodendrocytes	CNS		

		Make myelin in CNS — one cell wraps multiple axons	Slowed/lost saltatory conduction in CNS (multiple sclerosis)
Microglia	CNS	Resident immune / phagocytic cells; clear debris	Impaired CNS immune response and clearance
Ependymal cells	CNS — line ventricles & central canal	Make and circulate CSF (choroid plexus)	↓ CSF production or obstruction → hydrocephalus
Schwann cells	PNS	Make myelin in PNS — one cell wraps one axon segment	PNS demyelination (Guillain-Barré, Charcot-Marie-Tooth)
Satellite cells	PNS — in ganglia	Surround and support PNS neuron cell bodies	Ganglion dysfunction

★ **EXAM TIP** The **CNS/PNS myelin split**: "Oligo in One (CNS) — many axons. Schwann in Segments (PNS) — one axon."

This single distinction underlies the difference between MS (CNS, oligo) and Guillain-Barré (PNS, Schwann).

1.2 Conduction & Demyelinating Disorders

Stimulus → Na⁺ influx at node → action potential → impulse "jumps" from node to node along myelin → rapid signal arrival at synapse

WHY Why demyelination slows conduction. Without myelin between nodes, the axon must depolarize the entire length of membrane like an unmyelinated fiber. The signal arrives late, weakly, or not at all. Voluntary movement and sensation become delayed, fatigue with repetition (because Na⁺ channel reserves run out), and may stop entirely as disease progresses.

Disorder	Where	Cell affected	Classic presentation
Multiple sclerosis (MS)	CNS	Oligodendrocytes — autoimmune T-cell attack on CNS myelin	Young adult, female, "dissemination in time and space" — optic neuritis, weakness, sensory loss, relapsing
Guillain-Barré syndrome (GBS)	PNS	Schwann cells — autoimmune demyelination, often post-viral	Ascending symmetric flaccid weakness 1–3 weeks after a viral or GI infection (Campylobacter); CSF shows ↑ protein with normal cells
Charcot-Marie-Tooth	PNS	Hereditary demyelinating neuropathy	Childhood-onset distal weakness, "stork leg" deformity, foot drop

PATHOLOGY Guillain-Barré is the most common cause of acute flaccid paralysis in developed countries. The classic story: a young adult had a GI bug 1–3 weeks ago; now they have ascending symmetric weakness, areflexia, and tingling. Watch the diaphragm — respiratory failure is the killer. Treat with IVIG or plasmapheresis.

⚠ **COMMON MISSTEP** "MS and Guillain-Barré are the same demyelinating disease." No — completely different. MS = CNS, oligodendrocytes, relapsing-remitting, lifelong. GBS = PNS, Schwann cells, single ascending paralysis episode, often resolves. The exam loves making you pick between these two.

Practice Questions — Cells & Conduction

Q1. A patient with a CNS demyelinating disorder develops slowed nerve conduction. Which cell is most likely dysfunctional?

- A. Astrocyte
- B. Microglia
- C. Oligodendrocyte
- D. Schwann cell

[Show answer](#)

C. CNS myelin is made by oligodendrocytes. Schwann cells make PNS myelin. Astrocytes form the BBB; microglia are immune cells.

Q2. A patient has demyelination and develops slowed voluntary movement. The pathophysiology is best explained by:

- A. Loss of saltatory conduction along the axon
- B. Failure of neurotransmitter release at the synapse
- C. Reduced cerebral blood flow
- D. Loss of dendritic branching

[Show answer](#)

A. Demyelination disrupts the node-to-node "jumping" of action potentials. The signal still propagates but slowly and weakly, producing delayed and weak movements.

Q3. A patient develops ascending symmetric weakness 2 weeks after a viral GI illness. Which mechanism?

- A. CNS oligodendrocyte loss (multiple sclerosis)
- B. Peripheral nerve demyelination from autoimmune attack on Schwann cells (Guillain-Barré)
- C. Dopamine deficiency in basal ganglia
- D. Anterior horn cell death (poliomyelitis)

[Show answer](#)

B. Post-infectious ascending flaccid paralysis = Guillain-Barré. Schwann cells in PNS are the target. Treat with IVIG or plasmapheresis; monitor diaphragm.

Q4. Matching. Match each support cell to the dysfunction it causes when damaged.

Cell	Effect of damage
Astrocyte	—
Oligodendrocyte	—
Schwann cell	—
Ependymal cell	—

Choices: PNS demyelination · CNS demyelination · Loss of CSF circulation · Weakened blood-brain barrier

[Show answer](#)

Astrocyte → Weakened BBB · Oligodendrocyte → CNS demyelination · Schwann → PNS demyelination · Ependymal → Loss of CSF circulation.

Q5. Dropdown. A patient with myelin damage develops slowed voluntary movement because [select: saltatory conduction / synaptic release / Na/K pump activity] is disrupted and impulses no longer leap rapidly between [select: dendrites / nodes of Ranvier / synapses].

[Show answer](#)

Saltatory conduction · nodes of Ranvier. Myelin lets action potentials skip from node to node — remove myelin and the cascade slows or fails.

Unit 2 • CNS Protection: Meninges, CSF & Vascular Spaces LO1

2.1 The Three Meningeal Layers and the Spaces Between Them

Skull → EPIDURAL space → Dura mater → SUBDURAL space → Arachnoid → SUBARACHNOID space (CSF) → Pia mater → Brain

Layer / Space	Contents	Vessels there
Epidural space	Potential space between skull and dura	Meningeal arteries (esp. middle meningeal)
Dura mater	Tough outer layer attached to skull	—
Subdural space	Between dura and arachnoid	Bridging veins
Arachnoid mater	Web-like middle layer	—
Subarachnoid space	Holds CSF	Cerebral arteries (Circle of Willis)
Pia mater	Thin layer hugging brain surface	—

CSF — production, circulation, function

MECHANISM CSF is produced by the **choroid plexus** (specialized ependymal cells) inside the ventricles. It circulates through the lateral → 3rd → 4th ventricles → into the subarachnoid space → reabsorbed by **arachnoid villi** back into venous sinuses. Functions: cushions brain, removes waste, delivers nutrients, regulates pressure.

2.2 The Three Hematomas — Where the Blood Pools

Type	Vessel torn	Mechanism	Classic clinical picture	CT shape
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Epidural hematoma	Middle meningeal artery	Skull fracture (often temporal bone)	Brief loss of consciousness → "lucid interval" minutes to hours → rapid deterioration, fixed pupil; expands fast (arterial)	Lens-shaped (biconvex), does not cross sutures
Subdural hematoma	Bridging veins	Sudden head movement (fall, MVA) or atrophy (elderly, alcoholic)	Slower onset over hours to weeks; gradual confusion, headache, drowsiness	Crescent-shaped , crosses sutures
Subarachnoid hemorrhage	Berry aneurysm rupture (often Circle of Willis)	Spontaneous or HTN	"Worst headache of my life" — thunderclap, neck stiffness, photophobia	Blood in subarachnoid space, around brainstem and basilar cisterns
Intracerebral hemorrhage	Small penetrating arteries	HTN, AVM, trauma	Focal neurologic deficits depending on lesion location; gradual stroke-like onset	Blood within brain parenchyma

ACRONYM / MNEMONIC "E-A-V-S" for hematoma vessel and timing:

Epidural — **A**rterial — **f**ast (lucid interval)

Subdural — **V**enous — **s**low

Subarachnoid — aneurysm — sudden "worst headache"

PATHOLOGY The **lucid interval** is the classic epidural hematoma story. Patient is hit on the temple, briefly loses consciousness, "wakes up" feeling fine, then deteriorates rapidly as the arterial bleed expands. This is a neurosurgical emergency — drain the hematoma before brainstem herniation occurs.

2.3 Hydrocephalus

MECHANISM Hydrocephalus = excess CSF in the ventricles. Classified by where the problem is:

- **Communicating** (non-obstructive) — CSF flows through ventricles freely but is not *reabsorbed* at arachnoid villi (post-meningitis, post-subarachnoid bleed, congenital)
- **Non-communicating** (obstructive) — flow is *blocked* somewhere in the ventricular system. Classic site: **cerebral aqueduct** (aqueduct of Sylvius) between 3rd and 4th ventricles → CSF backs up into the lateral and 3rd ventricles → they enlarge.
- **Normal pressure hydrocephalus (NPH)** — adult chronic form: *Wet, Wobbly, Wacky* — incontinence, gait apraxia, dementia.

WHY Why **aqueduct obstruction enlarges ventricles upstream**. CSF made in the lateral ventricles must funnel through a narrow channel — the cerebral aqueduct — to reach the

4th ventricle. Block that channel, and CSF accumulates behind it. The lateral and 3rd ventricles balloon, pressing on surrounding brain tissue.

★ **EXAM TIP Hydrocephalus classic pictures:**

- Infant with rapidly enlarging head, sunset eyes, bulging fontanelle → congenital obstructive
- Older adult with the "3 W's" (Wet, Wobbly, Wacky) → NPH
- Adult after meningitis or subarachnoid bleed → communicating

Practice Questions — CNS Protection

Q6. A patient has a skull fracture and rapidly rising pressure between the skull and dura mater. Which mechanism explains this lesion?

- A. A meningeal artery is torn within the epidural space
- B. A bridging vein is torn within the subdural space
- C. A berry aneurysm has ruptured into the subarachnoid space
- D. CSF accumulates due to ependymal damage

[Show answer](#)

A. Skull fracture + rapid pressure between skull and dura = epidural hematoma. The middle meningeal artery is the most common source.

Q7. An elderly patient on anticoagulation falls and develops gradually worsening confusion over two weeks. CT shows a crescent-shaped collection along the cerebral convexity. Which lesion?

- A. Epidural hematoma
- B. Subdural hematoma — bridging vein rupture
- C. Subarachnoid hemorrhage
- D. Intracerebral hemorrhage

[Show answer](#)

B. Slow onset, crescent shape, bridging-vein source, elderly with brain atrophy that stretches those veins. Anticoagulation worsens the bleeding.

Q8. A patient develops ventricular enlargement after obstruction of the cerebral aqueduct. Which mechanism accounts for this change?

- A. CSF accumulates because normal fluid drainage is blocked
- B. Blood accumulates because arterial inflow is blocked
- C. Lymph accumulates because venous outflow is blocked
- D. Ions accumulate because neural signaling is blocked

[Show answer](#)

A. Aqueduct of Sylvius blockage prevents CSF flowing from 3rd to 4th ventricle → lateral and 3rd ventricles balloon → non-communicating (obstructive) hydrocephalus.

Q9. A 30-year-old presents with a "thunderclap headache" — the worst of her life — followed by neck stiffness. Which lesion is most likely?

- A. Epidural hematoma
- B. Subdural hematoma

- C. Subarachnoid hemorrhage from ruptured berry aneurysm
- D. Intracerebral hemorrhage

Show answer

C. "Worst headache of my life" + nuchal rigidity = subarachnoid hemorrhage until proven otherwise. CT first; if negative and suspicion remains, lumbar puncture for xanthochromia.

Unit 3 • Sensory Function & Pain LO2

3.1 Sensory Pathway — From Receptor to Perception

Skin receptor → peripheral nerve → dorsal horn of spinal cord → ascending tract → thalamus → primary sensory cortex (parietal lobe) → conscious perception

MECHANISM A signal must travel a four-neuron chain to be perceived. Cut any link — peripheral nerve, dorsal column, thalamus, cortex — and perception fails. A patient who is touched but does not feel it has a break somewhere on this pathway.

The two main ascending sensory tracts

Tract	Carries	Decussates (crosses)	Lesion → loss
Dorsal column / medial lemniscus	Fine touch, vibration, proprioception, 2-point discrimination	In the medulla (high)	Same side below the lesion if cord lesion; opposite side if brainstem lesion
Spinothalamic tract	Pain, temperature, crude touch	Within 1–2 levels of entering the cord (low)	Opposite side below the lesion

★ **EXAM TIP** "Pain crosses early, vibration crosses late." Spinothalamic = pain/temp = decussates near entry. Dorsal columns = vibration/proprioception = decussates in the medulla. This explains why hemicord lesions (Brown-Séquard) cause pain loss *opposite* the lesion and proprioception loss *same* side.

3.2 Pain Types

Type	Cause	Quality	Examples
Nociceptive — somatic	Tissue injury detected by nociceptors in skin, muscle, bone, joint	Sharp, throbbing, well-localized	Cut, fracture, sprain, surgical incision
Nociceptive — visceral	Stretching/distension of organs	Dull, achy, poorly localized; may refer (e.g., diaphragm pain to shoulder)	Appendicitis, MI (referred to arm/jaw), bowel obstruction
Neuropathic			Diabetic neuropathy, post-herpetic neuralgia, sciatica,

	Damage to peripheral or central nerves directly	Burning, shooting, electric, tingling, numb	trigeminal neuralgia, phantom limb
Referred	Visceral pain perceived at a distant somatic site (shared spinal level)	Same as visceral	MI → left arm/jaw; gallbladder → right shoulder; appendicitis → periumbilical

⚠ **COMMON MISSTEP** "Burning, shooting pain without a new injury" is the classic **neuropathic pain stem**. Don't pick nociceptive when you see those words. Neuropathic pain comes from the nerve itself being damaged or sensitized — diabetes, herpes, nerve compression, amputation.

WHY Why neuropathic pain doesn't respond well to opioids. Opioids dampen nociceptive signaling at the dorsal horn and brain. Neuropathic pain is generated by damaged neurons firing abnormally, often above the level the opioid acts. That's why we use anticonvulsants (gabapentin, pregabalin) and antidepressants (TCAs, SNRIs) — they calm hyperexcitable nerves.

Practice Questions — Sensory & Pain

Q10. A patient feels a pinprick at the skin but does not perceive it after a spinal cord lesion. Which best explains the loss of perception?

- A. The skin receptors are damaged
- B. The signal cannot ascend through the spinal cord to the cortex
- C. The motor pathway is interrupted
- D. The cerebellum is dysfunctional

[Show answer](#)

B. Receptor activation is intact (the patient feels it locally), but the ascending sensory tract is interrupted, so the signal cannot reach the cortex for conscious perception.

Q11. A patient describes burning, shooting pain in the foot without any new injury. Which type of pain is most likely?

- A. Visceral nociceptive pain
- B. Somatic nociceptive pain
- C. Neuropathic pain
- D. Referred pain

[Show answer](#)

C. "Burning, shooting" + no fresh injury = damaged nerve firing abnormally → neuropathic. Common causes: diabetes, post-herpetic neuralgia, sciatica.

Q12. Which finding is most consistent with nociceptive pain?

- A. Burning sensation along a dermatome with no injury
- B. Sharp, well-localized pain at a surgical incision
- C. Numbness and tingling in the toes from years of diabetes
- D. Phantom limb sensation after amputation

Show answer

B. Sharp, localized, tied to tissue injury = somatic nociceptive. The other options describe neuropathic pain.

Q13. Matching. Match each lesion location to its expected sensory deficit.

Lesion	Deficit
Peripheral nerve	—
Spinothalamic tract (cord)	—
Dorsal column (cord)	—
Primary sensory cortex	—

Choices: Loss of fine touch, vibration, proprioception same side · Loss of all modalities in nerve distribution · Loss of pain and temp opposite side · Cortical neglect or loss of fine discrimination opposite side

Show answer

Peripheral nerve → all modalities in nerve territory · **Spinothalamic** → pain/temp opposite side · **Dorsal column** → fine touch/vibration/proprioception same side · **Cortex** → cortical neglect / fine discrimination opposite side.

Unit 4 • Sleep, Arousal & Consciousness LO2

MECHANISM Arousal (being awake) is driven by the **reticular activating system (RAS)** — a network of neurons in the brainstem (especially the upper pons and midbrain reticular formation) that projects up to the thalamus and cortex. Without RAS input, the cortex cannot be aroused even if it is anatomically intact.

WHY Why brainstem injuries cause coma. A patient with a normal cortex but a destroyed RAS cannot be awakened — the cortex needs continual "wake-up" signals from the brainstem. This is why even small brainstem strokes can produce profound consciousness deficits, while large cortical strokes may leave the patient awake but with focal deficits.

Sleep architecture

Stage	Features	Regulator
NREM 1	Light sleep, easy to wake	Hypothalamus dampens RAS
NREM 2	Sleep spindles, K complexes	—
NREM 3 (slow-wave)	Deep sleep; growth hormone release; "physical restoration"	—
REM	Rapid eye movements, dreaming, atonia of voluntary muscles, "memory consolidation"	Brainstem (pons)

PATHOLOGY Sleep-wake cycle disruption. The suprachiasmatic nucleus (SCN) of the hypothalamus is the body's circadian pacemaker. The brainstem coordinates the sleep

stages themselves. Brainstem injury disrupts both REM and arousal — patients lose normal sleep architecture and may fluctuate between hypersomnia and unrousable states.

★ **EXAM TIP** "Difficult to arouse after brainstem injury" = damaged RAS. The level of consciousness is the responsibility of the brainstem, not the cortex.

Practice Questions — Sleep & Arousal

Q14. A patient becomes difficult to arouse after a brainstem injury. Which system is impaired?

- A. Reticular activating system (RAS)
- B. Cerebellum
- C. Hippocampus
- D. Basal ganglia

[Show answer](#)

A. RAS sits in the brainstem and drives cortical arousal. Damage there → impaired wakefulness even when the cortex is intact.

Q15. A patient develops disrupted sleep-wake cycles after a brainstem injury. Which structures most likely caused this?

- A. Hippocampus and amygdala
- B. Reticular activating system and pontine sleep regulators
- C. Cerebellum and vestibular nuclei
- D. Basal ganglia

[Show answer](#)

B. The brainstem houses both the RAS (arousal) and the pontine centers that regulate REM/NREM cycling. Injury here scrambles the architecture.

Unit 5 • Motor System & Movement Disorders LO3

5.1 The Two-Neuron Motor Pathway: UMN vs LMN

Motor cortex (UMN) → corticospinal tract → anterior horn cell (LMN starts) → peripheral nerve → neuromuscular junction → muscle

MECHANISM The motor system uses **two neurons in series**:

- **Upper motor neuron (UMN)** — cell body in the cortex; axon descends in the corticospinal tract through internal capsule and brainstem; ends in the spinal cord.
- **Lower motor neuron (LMN)** — cell body in the anterior (ventral) horn of the spinal cord; axon exits through ventral roots and peripheral nerves to reach muscle.

The *signs* of a lesion depend entirely on which neuron is broken.

Sign	UMN lesion	LMN lesion
Weakness	Yes (often hemiparesis)	Yes (focal, in nerve distribution)
Tone	Increased (spasticity)	Decreased (flaccid)

Reflexes	Hyperreflexic , clonus	Hyporeflexic or absent
Babinski	Positive (toes up)	Negative
Atrophy	Mild, late (disuse)	Marked, early
Fasciculations	No	Yes
Examples	Stroke, MS, spinal cord injury (chronic)	Polio, peripheral nerve injury, GBS, anterior horn disease

ACRONYM / MNEMONIC **UMN** = "EVERYTHING UP except strength." Tone up, reflexes up, toes up (Babinski).
LMN = "EVERYTHING DOWN." Tone down (flaccid), reflexes down, toes down. Plus visible atrophy and fasciculations.

5.2 Major Movement and Motor Disorders

Disorder	Lesion / Mechanism	Cardinal signs
Parkinson's disease	Loss of dopamine-producing neurons in substantia nigra → loss of basal ganglia inhibition → rigidity	TRAP : Tremor (resting), Rigidity (cogwheel), Akinesia/bradykinesia, Postural instability; mask-like face, shuffling gait
Amyotrophic lateral sclerosis (ALS / Lou Gehrig's)	Degeneration of both UMN and LMN	Mixed signs: spasticity AND atrophy/fasciculations; cognition spared; respiratory failure is the killer
Huntington disease	Autosomal dominant CAG repeat expansion → caudate atrophy	Mid-adult onset chorea, psychiatric symptoms, dementia
Cerebellar lesions	Damage to cerebellum (stroke, alcohol, MS)	Ataxia, intention tremor, dysmetria, dysdiadochokinesia, scanning speech, falls toward lesion side
Spinal cord injury — acute	Spinal shock — temporary loss of all reflex activity below the lesion	Flaccid paralysis , areflexia, autonomic instability for days to weeks
Spinal cord injury — chronic	Reflex arcs return below the lesion but UMN signal is still cut	Spasticity, hyperreflexia, clonus, possible muscle spasms

PATHOLOGY **Spinal shock to spasticity transition.** Right after spinal cord injury, even reflexes below the lesion vanish — flaccid paralysis. Over days to weeks, the spinal cord reflex arcs reorganize and start firing without UMN inhibition → the limbs become spastic, hyperreflexic, sometimes with painful spasms. Same lesion, two phases: acute = LMN-like; chronic = UMN-like.

⚠ **COMMON MISSTEP** "Parkinson's is a memory disease." No — Parkinson's is a *movement* disorder. Cognition can decline late, but the cardinal features are motor (TRAP). Alzheimer's is the classic memory disease.

★ **EXAM TIP** ALS is the only movement disorder with mixed UMN AND LMN signs. A patient with both spasticity and fasciculations, with normal cognition, points at ALS. This is a high-yield distinguishing feature.

Practice Questions — Motor

Q16. A patient with Parkinson's disease has slowed movement initiation, resting tremor, and cogwheel rigidity. Which mechanism best explains these findings?

- A. Excess dopamine in the cortex
- B. Loss of dopamine-producing neurons in the substantia nigra → reduced basal ganglia output
- C. Demyelination of the corticospinal tract
- D. Cerebellar degeneration

[Show answer](#)

B. Substantia nigra dopamine loss → impaired basal ganglia regulation of voluntary movement → rigidity, bradykinesia, resting tremor. Treatment with levodopa replenishes dopamine.

Q17. A patient develops flaccid paralysis with marked muscle atrophy and decreased reflexes in one arm. Which lesion?

- A. Upper motor neuron (corticospinal tract) lesion
- B. Lower motor neuron lesion
- C. Cerebellar hemisphere lesion
- D. Basal ganglia lesion

[Show answer](#)

B. Flaccid + atrophy + ↓ reflexes = LMN. Hyperreflexia and spasticity would suggest UMN.

Q18. A patient is diagnosed with ALS. Which combination of findings is most expected?

- A. Spasticity and hyperreflexia, but no muscle atrophy
- B. Flaccid paralysis without spasticity
- C. Both upper motor neuron signs (spasticity, hyperreflexia) and lower motor neuron signs (atrophy, fasciculations); cognition preserved
- D. Pure cerebellar ataxia

[Show answer](#)

C. ALS uniquely combines UMN + LMN signs because both neuron types degenerate. Cognition is typically spared (unlike many other neurodegenerative diseases).

Q19. A patient with a spinal cord injury initially has flaccid paralysis below the level of the injury. Several days later, the nurse notes increased muscle tone and intermittent spasms. Which best explains this transition?

- A. The injury has worsened
- B. Spinal shock has resolved; reflex arcs are returning without UMN inhibition, producing spasticity
- C. The patient is developing seizures
- D. The bones are healing

[Show answer](#)

B. Spinal shock = temporary areflexia after acute SCI. As reflex arcs reorganize without descending UMN control, spasticity, hyperreflexia, and clonus emerge.

Q20. A patient develops radiating leg weakness after an intervertebral disk ruptures into the spinal canal. Which mechanism?

- A. UMN compression in the cervical cord
- B. Compression of nerve roots (radiculopathy) → LMN-pattern weakness in the affected dermatome/myotome
- C. Cerebellar herniation
- D. Astrocyte activation

[Show answer](#)

B. Disk herniation compresses spinal nerve roots, producing LMN-type weakness, sensory deficits, and reflex loss along the affected nerve's distribution. Sciatica is the classic presentation.

Unit 6 • Cognition: Dementia, Delirium & Consciousness

LO3

MECHANISM Dementia = chronic, progressive, irreversible decline in cognition. Memory, language, executive function, and judgment slowly erode over months to years.

Delirium = acute, fluctuating, often *reversible* disturbance in attention and consciousness, usually caused by medical illness, drugs, or hospitalization. Hours to days.

Feature	Dementia	Delirium
Onset	Insidious (months to years)	Acute (hours to days)
Course	Slowly progressive	Fluctuating; classically worse at night ("sundowning")
Attention	Preserved early; declines late	Impaired from the start — defining feature
Consciousness	Clear	Altered (hyperalert or drowsy)
Reversibility	Usually no (with rare exceptions)	Usually yes when underlying cause is treated
Common causes	Alzheimer's, vascular, Lewy body, frontotemporal	Infection (UTI in elderly), medications (anticholinergics, opioids, benzos), metabolic derangement, alcohol withdrawal, post-op, ICU stay

★ **EXAM TIP** The single best discriminator: **onset**. If the change happened over a weekend, it's delirium. If it happened over months, it's dementia. A demented patient who suddenly worsens may have *delirium superimposed on dementia* — usually triggered by a UTI, medication, or pneumonia.

WHY Why elderly patients delirium-out so easily. Aging brains have less cognitive reserve. Add a medication, an infection, a hospital admission, or sleep deprivation, and the system tips into delirium. Treating the underlying cause and minimizing tethering, sedatives, and sensory deprivation usually reverses it.

Levels of consciousness — the spectrum

- **Alert** — fully aware, responsive
- **Confused** — disoriented, may follow simple commands
- **Lethargic** — drowsy but rouses to voice
- **Obtunded** — rouses to vigorous stimulation, slow response
- **Stuporous** — rouses only to painful stimuli, returns to unresponsive state
- **Comatose** — no purposeful response to any stimulus

⚠ **COMMON MISSTEP "A demented patient is just confused."** Don't blame all confusion on baseline dementia. A new behavior change in a demented patient is delirium until proven otherwise — search for infection, medication change, hypoxia, electrolyte abnormality, or pain.

Practice Questions — Cognition

Q21. Which feature most reliably distinguishes dementia from delirium?

- A. Memory loss
- B. Acute fluctuating course with impaired attention
- C. Age over 65
- D. Hallucinations

[Show answer](#)

B. Acute onset, fluctuating, and impaired attention define delirium. Memory loss can occur in both. Dementia is gradual and consciousness/attention are preserved early.

Q22. An 82-year-old with mild dementia is admitted with pneumonia and becomes acutely confused, agitated at night, and inattentive. Most likely cause?

- A. Worsening Alzheimer's
- B. Stroke
- C. Delirium superimposed on dementia, triggered by infection and hospitalization
- D. New psychiatric illness

[Show answer](#)

C. Acute change + inattention + hospitalized + infection = delirium on top of dementia. Treat the pneumonia, minimize tethers and sedatives, restore sleep-wake cycle, and the delirium typically resolves.

Unit 7 • Cerebral Hemodynamics & Intracranial Pressure

LO3

7.1 The Monro-Kellie Doctrine

MECHANISM The skull is a fixed compartment containing three things: **brain tissue (~80%)**, **blood (~10%)**, **CSF (~10%)**. If one expands, the others must shrink to keep total volume — and pressure — constant. When compensation is exhausted, ICP rises sharply.

Mass added (tumor, blood, edema) → CSF first displaced → blood volume reduced → compensation exhausted → ICP rises rapidly → herniation

Normal ICP: 5–15 mm Hg. Sustained > 20 mm Hg is pathologic.

7.2 Cerebral Autoregulation

MECHANISM Cerebral blood vessels constrict or dilate to keep cerebral blood flow constant despite changes in MAP. Two key chemical influences:

- ↑ **CO₂** (or ↓ **pH**) → cerebral **vasodilation** → ↑ blood flow → ↑ ICP
- ↓ **CO₂** → cerebral **vasoconstriction** → ↓ blood flow → ↓ ICP (this is why we briefly hyperventilate patients with herniation)
- ↓ **O₂** → cerebral **vasodilation**

WHY Why a **CO₂** rise is dangerous in head injury. Hypoventilation → ↑ PaCO₂ → cerebral vasodilation → more blood in the skull → ↑ ICP → less cerebral perfusion. Patients with raised ICP are kept normoventilated (PaCO₂ around 35–40 mm Hg) and oxygenated to avoid this trap.

7.3 Cushing Triad — The Late Warning

PATHOLOGY When ICP rises so high that the brainstem is being compressed, the body responds with the **Cushing reflex** — a sympathetic surge to push blood through the brain:

- **Hypertension** (↑ systolic BP, widening pulse pressure)
- **Bradycardia**
- **Irregular respirations** (Cheyne-Stokes or apneic)

This is a late, ominous sign — herniation is imminent.

7.4 Early vs Late Signs of Rising ICP

Early	Late
Headache, restlessness, irritability	Decreasing level of consciousness
Subtle confusion, change in behavior	Cushing triad
Pupils still reactive	Fixed, dilated pupil (often unilateral first — "blown pupil")
Vomiting (often projectile)	Posturing (decorticate or decerebrate)
	Coma, herniation, apnea

Visual changes, papilledema on exam

★ **EXAM TIP** "A head-injured patient who becomes restless and confused" is showing an **EARLY** sign of rising ICP — not just psychological distress. Behavior change before posturing is when the team has time to act.

⚠ **COMMON MISSTEP** "Hypotension and tachycardia mean rising ICP." Wrong direction. Cushing triad is **HTN + bradycardia + irregular respirations**. Hypotension + tachycardia points to hypovolemia, sepsis, or shock.

Practice Questions — Cerebral Hemodynamics & ICP

Q23. A patient with rising carbon dioxide develops compensatory cerebral vasodilation. Which mechanism best explains this response?

- A. CO₂ acts as a vasoconstrictor
- B. ↑ CO₂ lowers pH locally → cerebral vessels dilate to increase blood flow and CO₂ clearance
- C. CO₂ causes anaphylaxis
- D. The kidney releases vasodilators

[Show answer](#)

B. CO₂ is a potent cerebral vasodilator. Hypercapnia → ↓ brain tissue pH → arteriolar dilation → ↑ flow. This is autoregulation working — but in a head-injured patient with already-high ICP, it can be dangerous.

Q24. A patient with a head injury becomes increasingly restless and confused. Which finding is MOST concerning for worsening intracranial pressure?

- A. Anxiety alone
- B. New-onset hypertension with bradycardia and irregular respirations
- C. Heart rate of 95
- D. Mild headache

[Show answer](#)

B. Cushing triad — HTN, bradycardia, irregular respirations — is the late warning of severe ICP elevation and impending herniation. Restlessness/confusion is an earlier sign that should still trigger urgent assessment.

Q25. Select all that apply. A patient develops rising PaCO₂. Which cerebral hemodynamic responses may occur?

- A. Cerebral vasodilation
- B. Increased cerebral blood flow
- C. Increased intracranial pressure
- D. Cerebral vasoconstriction
- E. Decreased intracranial pressure

[Show answer](#)

A, B, C. Hypercapnia dilates cerebral vessels → more blood in the skull → higher ICP. D and E describe the response to *hypocapnia* (low CO₂), which is the opposite.

Appendix • Top Missteps & Master Mnemonics

Top 15 Week 4 Missteps

#	The trap	What's actually true
1	"Schwann cells make CNS myelin."	Schwann = PNS. Oligodendrocytes = CNS.
2	"MS and Guillain-Barré are the same disease."	MS = CNS, oligodendrocytes, relapsing. GBS = PNS, Schwann cells, ascending, single episode.
3	"Epidural hematoma is venous and slow."	Epidural = arterial (middle meningeal), fast , lucid interval. Subdural = venous (bridging veins), slow.
4	"Subarachnoid hemorrhage causes a mild headache."	It's the " worst headache of my life " — thunderclap. With nuchal rigidity.
5	"All hydrocephalus is from too much CSF production."	Most is from impaired flow or absorption , not overproduction.
6	"Burning, shooting pain is nociceptive."	That's neuropathic . Nociceptive is sharp/throbbing/well-localized.
7	"Brainstem injury causes only motor problems."	It also wipes out arousal (RAS) and sleep architecture — coma and disrupted sleep-wake cycles.
8	"Parkinson's is a memory disease."	Movement disorder. TRAP: tremor, rigidity, akinesia, postural instability.
9	"ALS spares lower motor neurons."	ALS hits both UMN and LMN — that's its signature.
10	"Spinal shock is permanent flaccid paralysis."	It's a temporary phase . Spasticity emerges as reflex arcs reorganize.
11	"Delirium is permanent."	Delirium is acute and usually reversible when the cause is treated.
12	"All confusion in the elderly is dementia."	Acute change = delirium until proven otherwise. Look for infection, drugs, electrolytes.
13	"Cushing triad is HTN + tachycardia."	It's HTN + bradycardia + irregular respirations — late ICP elevation.
14	"Low CO ₂ raises ICP."	Low CO ₂ → vasoconstriction → lowers ICP. High CO ₂ raises it.
15	"Pain crosses the cord at the brainstem."	Spinothalamic pain crosses at entry (low). Dorsal column proprioception crosses at the medulla (high).

Master Mnemonic Sheet

Mnemonic	Meaning
"O in CNS, S in PNS"	Oligodendrocytes make CNS myelin; Schwann make PNS myelin
TRAP	Parkinson's: Tremor, Rigidity, Akinesia, Postural instability

UMN = "EVERYTHING UP"	Tone up, reflexes up, toes up (Babinski). Atrophy late, no fasciculations.
LMN = "EVERYTHING DOWN"	Tone down (flaccid), reflexes down. Atrophy early, fasciculations present.
"Pain crosses early, vibration crosses late"	Spinothalamic decussates at cord entry; dorsal columns at medulla
"E-A-V-S"	Epidural=Artery, subDural=Vein, Subarachnoid=aneurysm
"Lens vs Crescent"	Epidural is lens-shaped (does not cross sutures); subdural is crescent-shaped (crosses sutures)
3 W's	Normal pressure hydrocephalus: Wet (incontinent), Wobbly (gait), Wacky (dementia)
"Worst headache of my life"	Subarachnoid hemorrhage
Cushing triad	↑ BP + ↓ HR + irregular respirations = late, severe ICP elevation
Acute = Delirium, Chronic = Dementia	Onset is the single best discriminator
"ALS = both"	UMN signs + LMN signs together, cognition spared

Quick Reference — Lesion Localization

Symptom	Most likely location
Slowed conduction in CNS	Oligodendrocyte injury (e.g., MS plaques)
Ascending flaccid paralysis post-virus	PNS Schwann cells (Guillain-Barré)
Lucid interval after head trauma	Epidural hematoma (middle meningeal artery)
Crescent CT in elderly faller	Subdural hematoma (bridging veins)
Resting tremor + rigidity	Substantia nigra → basal ganglia
Mixed UMN + LMN signs	ALS
Difficult to arouse	RAS / brainstem
Acute attention disturbance	Delirium
HTN + bradycardia + irregular respirations	Severe ICP elevation (Cushing triad)

Week 4 Study Strategy

- **Master the support cells first.** Half of the LO1 questions are decided by knowing which glial cell does what.
- **Anchor every disorder to its location.** CNS or PNS? Cortex, brainstem, basal ganglia, cerebellum, cord, root, peripheral nerve, neuromuscular junction, or muscle? Location dictates symptoms.
- **Drill UMN vs LMN sign tables.** Many motor questions reduce to which side of the dotted line you're on.
- **Memorize the three hematomas as a set.** Vessel, space, timing, and CT shape — never one without the others.

- **Know the Cushing triad cold.** It's the highest-yield late-ICP teaching point and easy points if you have it.
- **Onset over hours = delirium. Onset over months = dementia.** Don't overthink it.
- **Spend extra time on the sensory-tract crossings.** Brown-Séquard hemicord lesion is a perennial favorite.

SCI225 Week 4 Study Guide — built from the Week 4 learning objectives covering neurologic structure & function, sensory and pain, sleep, motor function, cognition, and cerebral hemodynamics. Made by Dr. Vidya Aggrawal to teach the concepts tested, not the exam questions themselves. Same color-coded format as the Week 1 and Week 2 guides. Print for paper, or use in-browser with collapsible practice answers.